AA–TV wiring v8.1: no wheels. 6x Deutsch DTP 4–pin (wet zone), servo leads + cables epoxy–potted into dryboxes, 14 AWG marine outside / 20 AWG on boards.

BUOY (surface, dry)

5V_PI
BT2
USB–C power bank 5 V / 3 A
GND_PI

J1

GPIO12/PWM0		5V	2 — 5V_PI
GPIO13/PWM1		3V3	1
GPIO14/UART_TXD		ID_SD_I2C0/GPIO00	27
GPIO15/UART_RXD		ID_SC_I2C0/GPIO01	28
GPIO16/SPI1_CE2		SDA_I2C1/GPIO02	3
GPIO17/SPI1_CE1		SCL_I2C1/GPIO03	5
GPIO18/SPI1_CE0/PCM_CLK/PWM0			
GPIO19/SPI1_MISO/PCM_FS		GPCLK0/GPIO04	7
GPIO20/SPI1_MOSI/PCM_DIN/PWM1		GPCLK1/GPIO05	29
GPIO21/SPI1_SCLK/PCM_DOUT		GPCLK2/GPIO06	31
GPIO22/SDIO_CLK		CE1_SPI0/GPIO07	26
GPIO23/SDIO_CMD		CE0_SPI0/GPIO08	24
GPIO24/SDIO_DAT0		MISO_SPI0/GPIO09	21
GPIO25/SDIO_DAT1		MOSI_SPI0/GPIO10	19
GPIO26/SDIO_DAT2		SCLK_SPI0/GPIO11	23
GPIO27/SDIO_DAT3			
GND			6 — GND_PI

Raspberry Pi 4 (buoy)

Pi 4 talks to the Pico over USB serial (/dev/ttyACM0) and powers it from USB 5 V. Reset the Pico by power–cycling the Pi USB ports (uhubctl).

TETHER

W1
USB–A — micro–B
+ USB data
5V_PI — VBUS_PICO
GND_PI — GND

USB–A to micro–B, ~3 m

Vinyl air tube from the buoy; USB cable runs inside it. USB plugs are dry (buoy, Pico).

PEN1
A1: air–tube fitting (USB inside)

MAIN DRYBOX (upper, on hull)

BOARD A: Pico solder–board

Raspberry Pi Pico
A1

GP12	R1 220	M4_PWM	
GP13	R2 220	M5_PWM	
GP14	R3 220	M6_PWM	
GP15	R4 220	M7_PWM	
GP16	R5 220	THR_PWM	
GP19	R6 220	PUMP_PWM	

VBUS_PICO, +3V3, VSYS, VBUS, 3V3
RUN, 3V3_EN
GPIO0 … GPIO15
GPIO16 — GP16
GPIO17
GPIO18
GPIO19 — GP19
GPIO20
GPIO21
GPIO22
ADC_VREF
GPIO26_ADC0 — LEAK_MAIN
GPIO27_ADC1 — LEAK_REAR
GPIO28_ADC2
AGND — GND
GND

220 ohm in series on every PWM line that leaves the drybox.

TA2
1 — +3V3
2 — GND
3 — LEAK_MAIN
Main leak sensor

TA1

M4_PWM	1	1	M4_PWM
M5_PWM	2	2	M5_PWM
M6_PWM	3	3	M6_PWM
M7_PWM	4	4	M7_PWM
THR_PWM	5	5	THR_PWM
H1_SPARE6	6	6	H1_SPARE6
H1_SPARE7	7	7	H1_SPARE7
PUMP_PWM	8	8	PUMP_PWM
LEAK_REAR	9	9	LEAK_REAR
+3V3	10	10	+3V3
GND	11	11	GND
GND	12	12	GND

H1 out (14 AWG)

All Board A terminals: 5.08 mm pitch (0.2 in, fits the breadboard grid). TA1 pins 6–7: spare H1 conductors.

PEN2
A2: H1 cable, epoxy–potted

REAR DRYBOX (inside open rear endcap)

BOARD B: servo / signal board

TB1
M4_PWM 1
M5_PWM 2
M6_PWM 3
M7_PWM 4
THR_PWM 5
H1_SPARE6 6
H1_SPARE7 7
PUMP_PWM 8
LEAK_REAR 9
+3V3 10
GND 11
GND 12
H1 in (14 AWG)

TB7
1 — SERVO_7V5
2 — GND
7.5 V in (14 AWG)
SERVO_7V5
C1 1000uF 16V
GND

Board B:
– 7.5 V + GND rails: strip of 14 AWG soldered along strips
– Servo + / – jumpers: 14 AWG
– Signal jumpers: 20 AWG
– No 12 V on the breadboard
– TB1 pins 6–7: spare, unused

TB2
M4_PWM 1
SERVO_7V5 2
GND 3
M5_PWM 4
SERVO_7V5 5
GND 6
Rudder servo leads (potted)

TB3
M6_PWM 1
SERVO_7V5 2
GND 3
M7_PWM 4
SERVO_7V5 5
GND 6
Elevator servo leads (potted)

TB4
THR_PWM 1
GND 2
Thruster ESC signal (to D3)

TB5
PUMP_PWM 1
GND 2
Pump ESC signal (U3)

TB6
+3V3 1
GND 2
LEAK_REAR 3
Rear leak sensor

PEN3
R2: signal cables potted (H1, 4 servo leads, THR sig)

Off–board, in rear drybox

BB1
VBAT
Bus bars (screw studs, ring terminals):
in: D1 trunk, one 14 AWG per pole
out: buck U1, pump ESC U3, D3 thruster ESC feed (1 stud spare)
+12 V bus bar (>=30 A)

BB2
GND
GND bus bar (>=30 A)

U1 BUCK
VBAT — VIN+
GND — VIN–
VOUT+ — SERVO_7V5
VOUT– — GND
12 V –> 7.5 V buck (>=5 A)

U3
VBAT — BAT+
GND — BAT–
BEC+
SIG
SIG_GND
M+ — PUMP_M+
M– — PUMP_M–
PUMP_PWM — ESC
GND
Pump ESC (brushed, reversible)
Pump ESC silts dry, so any brushed ESC with reverse works. BEC wire cut. Motor leads –> R3 –> D2 pins 1–2 (extend with 14 AWG if short).

PEN4
R3: power cables potted (trunk, thruster feed, pump)

WET ZONE (flooded hull, open rear endcap, splash areas): Deutsch DTP 4–pin connectors live here

20 A ATC, waterproof holder
F1
VBAT
BT1
3S LiPo (wet, no box)
SW1
GND
Kill switch, low side, >=30 A

D1A D1B
VBAT 1 — 1 VBAT
GND 2 — 2 GND
D1_LOOP 3 — 3 D1_LOOP
D1_LOOP 4 — 4 D1_LOOP
D1 battery DTP06–4S DTP04–4P
D1: 1 + / 2 GND (25 A each); 3–4 dummy loop.

Battery side: battery + –> F1 –> D1 pin 1; battery – –> SW1 –> D1 pin 2. D1 pins 3–4: dummy loop on each half (short 14 AWG piece, contact on each end) so every cavity is sealed. No splits. Charge through a DTP04–4P –> XT60 adapter.

D4A D4B
M4_PWM 1 — 1 M4_PWM
M5_PWM 2 — 2 M5_PWM
M6_PWM 3 — 3 M6_PWM
M7_PWM 4 — 4 M7_PWM
D4 H1 1–4 DTP06–4S DTP04–4P

D5A D5B
THR_PWM 1 — 1 THR_PWM
H1_SPARE6 2 — 2 H1_SPARE6
H1_SPARE7 3 — 3 H1_SPARE7
PUMP_PWM 4 — 4 PUMP_PWM
D5 H1 5–8 DTP06–4S DTP04–4P

D6A D6B
LEAK_REAR 1 — 1 LEAK_REAR
+3V3 2 — 2 +3V3
GND 3 — 3 GND
GND 4 — 4 GND
D6 H1 9–12 DTP06–4S DTP04–4P
H1 main <–> rear: 12 x 14 AWG over D4, D5, D6. Pins 8–7 spare.

M4_PWM 1 — SERVO_7V5 2 — GND 3 — M5
Rudder servo
M5_PWM 1 — SERVO_7V5 2 — GND 3 — M5
Rudder servo
M6_PWM 1 — SERVO_7V5 2 — GND 3 — M6
Elevator servo
M7_PWM 1 — SERVO_7V5 2 — GND 3 — M7
Elevator servo

Servo leads: potted straight into the rear drybox at R2, no connector. BACKUP if a lead can't reach the box: one MGYOO 20 AWG 3–pin waterproof connector per servo, inline in the wet zone.

D3A D3B
VBAT 1 — 1 VBAT
GND 2 — 2 GND
THR_PWM 3 — 3 THR_PWM
GND 4 — 4 GND
D3 thruster DTP06–4S DTP04–4P
D3: 1 +12 V, 2 GND, 3 PWM (yellow), 4 signal GND (white). All 14 AWG.

U2 BLDC ESC
VBAT — BAT+
GND — BAT–
THR_PWM — SIG
GND — SIG_GND
U V W — 1 2 3 — MT1 BLDC
TD1.2 thruster
TD1.2 ESC (waterproof, wet)
Factory thruster cable stays on the ESC.

D2A D2B
PUMP_M+ 1 — 1 PUMP_M+
PUMP_M– 2 — 2 PUMP_M–
D2_LOOP 3 — 3 D2_LOOP
D2_LOOP 4 — 4 D2_LOOP
D2 pump DTP06–4S DTP04–4P
D2: 1 pump +, 2 pump –; 3–4 dummy loop.

PUMP_M+
M3
Ballast pump, 12 V brushed (TBD, <=5 A)
PUMP_M–

Deutsch DTP: size 12 contacts, 12–14 AWG, 25 A each. Every cavity is filled: unused pins get a dummy wire loop. Source side (battery / drybox) gets the DTP06 plug with socket contacts.

Wire: 14 AWG marine for everything outside the boxes, every DTP contact and all power; 20 AWG hookup inside.
Only connectors: 6x Deutsch DTP 4–pin in the wet zone + 5.08 mm screw terminals / bus bars inside.
Two solder–boards (Board A main drybox, Board B rear drybox). Cables epoxy–potted through drybox walls.

PIE AA–TV
Sheet: /
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Title: AA–TV electrical v8.1: no wheels, 6x Deutsch DTP 4–pin, servos potted

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